

HARSH VARDHAN

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EDUCATION

JSS Academy of Technical Education Noida
Computer Science & Engineering (Data Science)

Bachelor of Technology
2022 - 2026

EXPERIENCE

Grand India | Data & ML Engineer

Remote | June 2025 - Sept 2025

- Built and deployed end-to-end ML pipelines for logistics optimization, reducing delivery time by 35%.
- Built predictive forecasting models for inventory management, reducing storage cost by 20%.
- Implemented CNN-based computer vision system achieving 95% accuracy for automated inspection.
- Built time-series forecasting models using statistical techniques for inventory prediction.

SKILLS

Programming Languages:	Python (NumPy, Pandas), SQL, Java
ML/AI:	Neural Networks, NLP, Computer Vision, RAG, LLM Orchestration (Lang-Graph), Model Evaluation, Feature Engineering
Frameworks:	Scikit-learn, TensorFlow, PyTorch, Keras, CatBoost
Data Systems:	Pinecone, Neo4j, Azure SQL, PostgreSQL, MongoDB ,AWS
Backend/Deployment:	FastAPI, REST APIs, AWS Lambda, Vercel, Render
Generative AI:	RAG, Prompt Engineering, LLM Evaluation, Embeddings, Vector Search
Core CS:	Data Structures & Algorithms, DBMS, Operating Systems, OOP, Computer Networks, System Design Basics

PROJECTS

Agentic Hybrid SQL RAG System | [GitHub](#) | [Live Demo](#)

Tech Stack : Python, LangGraph, Pinecone, Neo4j, Google Gemini, HuggingFace Inference API, FastAPI, SQLAlchemy, Azure SQL, Agentic AI

- Built a production-grade Text-to-SQL system over a 70-table Azure SQL (AdventureWorks2019) database, converting natural language queries into executable SQL with live results.
- Designed a LangGraph-based agentic pipeline with self-correcting retries using error feedback, improving SQL execution success rate from 60% → 90%.
- Implemented hybrid retrieval: Pinecone vector search (384-dim embeddings via HuggingFace API) + Neo4j graph-based join-path discovery, reducing incorrect joins by 35% – 45%.
- Built a production-grade Retrieval-Augmented Generation (RAG) system using embeddings, vector databases, and LLM orchestration

AI-powered financial intelligence platform | [GitHub](#) | [Live Demo](#)

Tech Stack : PyTorch, Python, FastAPI, Scikit-Learn, Pandas, WebSockets, yfinance, Finnhub API

- Developed a multi-step LSTM (Long Short-Term Memory) Neural Network using PyTorch to predict 15-period forward-looking stock prices based on a 60-day historical sequence.
- Engineered an input pipeline featuring 7 technical & market indicators (Open, High, Low, Close, Volume, 14-day RSI, and 20-day SMA) utilizing the ta library to capture momentum and trend signals.
- Formulated a recursive, autoregressive inference loop combined with customized volatility-based post-processing to generate realistic OHLC (Open-High-Low-Close) candle wicks and bodies based on trailing market variance.
- Built a high-performance FastAPI backend that feeds historical data scaling scalers and model weights, ensuring real-time multi-step predictions are generated and returned in under 100ms.

FinAlologica | [GitHub](#) | [Live Demo](#)

Tech Stack : PyTorch, Torchvision, PostgreSQL, Streamlit, Python, JavaScript, React, Node.js, Express.js, FastAPI

- Built a scalable full-stack ML application using React (Redux Toolkit), Node.js/Express, and a FastAPI microservice, enabling real-time inference with a modular, independently deployable architecture.
- Fine-tuned a MobileNetV3 (PyTorch) image classification model, achieving 95% accuracy for fish species identification, exposed via low-latency REST APIs.
- Developed a context-aware recommendation engine integrating real-time weather data (Open-Meteo API) with domain rules, improving recommendation relevance by 40%.
- Automated species tagging and fishing recommendations, reducing manual logging effort by 30% and improving overall user workflow efficiency.

CERTIFICATIONS

- ML, Deep Learning & GenAI 21 Projects, 21 Days - GeeksforGeeks | [Link](#)
- Google Analytics 360 - Google Analytics Academy.

HONORS & AWARDS

- **Smart India Hackathon 2023:** Runner-Up (Top 15 nationally) — Led 5-member team building India Tourism Platform with React.js and Node.js for government problem statement.
- Competed in the State-Level Taekwondo Championship, demonstrating high performance, focus, and athletic commitment.
- Solved 300+ DSA problems across LeetCode and coding platforms.